

Method / Variant

Perplexity (↓ better)

(1) Core mechanism

AdamW baseline

17.78 ▲

Remove second moment v_t (SGD + Momentum)

70.74

+ Newton-Schulz orthogonalization

16.86

+ LR scaling

16.76

+ Nesterov (full Muon)

16.60 ★

(2) Gain design

LR scaling: stronger one-way

16.60

LR scaling: conservative one-way

16.66

LR scaling: symmetric two-way

16.52 ★

LR scaling: penalize wide matrices

16.71

Nesterov after orthogonalization

16.57 ★

(3) Order constraints

Momentum in original-gradient space

16.60

Momentum in orthogonalized space

23.01

LR scaling after NS

16.60

LR scaling before NS

16.70

Decoupled weight decay

16.60

Coupled weight decay

16.60

16.0

16.5

17.0

17.5

18.0